Check for updates

Received: 19 November 2020 | Revised: 21 June 2021 | Accepted: 10 July 2021

DOI: 10.1002/ird.2637

WILEY

REVIEW ARTICLE

On-farm irrigation water management in India: Challenges and research gaps*

Ajita Gupta[1] | **Ranjay kumar Singh**[1] | **Mukesh Kumar**[1] | **Chetankumar Prakash Sawant**[1] | **Bhaskar Bharat Gaikwad**[2]

[1]ICAR-Central Institute of Agricultural Engineering, Bhopal, India

[2]ICAR-National Institute of Abiotic Stress Management, Baramati, India

Correspondence
Ajita Gupta, scientist, ICAR-Central Institute of Agricultural Engineering, Bhopal, India.
Email: ajitagupta2012@gmail.com

Abstract

The growing demand for food to feed the increasing population of India has made it necessary to bring more land under cultivation with assured irrigation facilities. The irrigation sector demands a high amount of the available freshwater resources of the country. Thus, the efficiency of its use is of utmost importance for sustainable agriculture. India is the second-largest irrigated country and is one of the leading countries in terms of the expansion of irrigated areas in the world. Researchers and various agencies are working towards increasing the irrigation efficiency of the system and the water productivity of the crops. The formulation of any irrigation scheme and policy to be sustainable should be based on scientific inputs and supported by pertinent research studies. The effective communication between policymakers and the research community is the main key to the success of such formulated schemes and policies. This paper provides an overview of current challenges and problems associated with irrigation at the field level vis-à-vis research gaps for scientific management of irrigation water in the farmer's field. The apathy of farmers for adopting modern irrigation practices has also been reasoned in this review paper. This review paper would provide an insight into common irrigation-related problems of farmers to researchers and policymakers.

KEYWORDS

automation, groundwater, India, irrigation schemes, microirrigation, surface irrigation

Résumé

La demande croissante de denrées alimentaires pour nourrir la population croissante de l'Inde a rendu nécessaire la mise en culture de plus de terres avec des installations d'irrigation assurées. Le secteur de l'irrigation consomme une énorme quantité de ressources en eau douce du pays. Par conséquent, l'efficacité de son utilisation est de la plus haute importance pour une agriculture durable. L'Inde a la deuxième plus grande superficie irriguée au monde après la Chine et est l'un des principaux pays dans l'expansion des zones irriguées. Les chercheurs et divers organismes travaillent à accroître l'efficacité

*Gestion de l'eau d'irrigation à la ferme en Inde: défis et lacunes de la recherche

 © 2021 John Wiley & Sons, Ltd.

de l'irrigation du système et l'efficacité de l'utilisation de l'eau des cultures. La formulation de tout programme et politique durable lié à l'irrigation doit être basée sur des preuves scientifiques et doit fortement être soutenu par des études de recherche pertinentes. L'inefficacité de la communication entre les décideurs politiques et la communauté des chercheurs est préjudiciable au succès de ces programmes et politiques formulés. Cet article donne un aperçu des défis et des problèmes actuels rencontrés sur le terrain ainsi que des questions de recherche pour la gestion scientifique de l'eau d'irrigation dans le champ de l'agriculteur. L'apathie des agriculteurs envers la non-adoption de méthodes modernes d'irrigation a également été motivée dans ce document de synthèse. Ce document de synthèse fournirait aux chercheurs et aux décideurs un aperçu des problèmes courants liés à l'irrigation des agriculteurs.

MOTS CLÉS

Automatisation, Eaux souterraines, Inde, Schémas d'irrigation, Microirrigation, Irrigation de surface

1 | INTRODUCTION

The agriculture sector is of prime importance to India's economy as it contributes 14%–15% to the country's gross domestic product and provides employment opportunities to over 55% of the population (Chand et al., 2017). Irrigation plays a pivotal role in the agricultural, social, and economic growth of a nation by providing stability to its food production and productivity. It is the most vital input, along with high-yielding varieties that led the green revolution in India. The Indian irrigation system got wheels of development during the colonial era, but extensive work started post independence in the 19th century (Namara et al., 2005). During the 20th century, with the advent of modern civil engineering equipment and machinery, series of synchronized institutional reforms and government policies such as the five-yearly economic plans encouraged the involvement of leading international development agencies, which enhanced irrigation development in India. Out of 329 Mha (million hectare) of total geographical area, India has the highest cultivated area of the world with a reported net sown area of 141 Mha and a gross cropped area of 201 Mha (Government of India [GoI], 2018). The estimated ultimate sustainable irrigation potential of India is about 140 Mha (Das, 2003); however, only 68 Mha of the cultivated area has been brought under irrigation, while most of the remaining area is still monsoon dependent (GoI, 2018). Despite these facts, India is still one of the leading countries in terms of the growth of irrigated area (Figure 1).

Of the current global cultivated area of 1,600 Mha, about 1,300 Mha (80%) is rainfed, which contributes about 60% of global crop output in a variety of production systems (Food and Agriculture Organization of the United Nations [FAO], 2017). Irrigated agriculture accounts for about 20% of the cultivated area worldwide, while it generates an estimated 40% of crop production (FAO, 2015). India ranks first among the rainfed countries in the world in terms of area (71.7 Mha) (National Rainfed Area Authority [NRAA], 2020) but ranks among the lowest in yields (around 1 t/ha) (Rao et al., 2015). The main crops in rainfed and dryland agriculture are coarse cereals (85%), pulses (83%), oilseeds (70%), and cotton (65%), contributing around 44% of total agricultural output (Nagaraj, 2013). Irrigation systems typically result in yields at least twice that of rainfed crops (Wani et al., 2009). Globally, rainfed cereal yields in the developing world average 1.5 t/ha, but irrigated yields are 3.3 t/ha (Rosegrant et al., 2002). However, in Latin

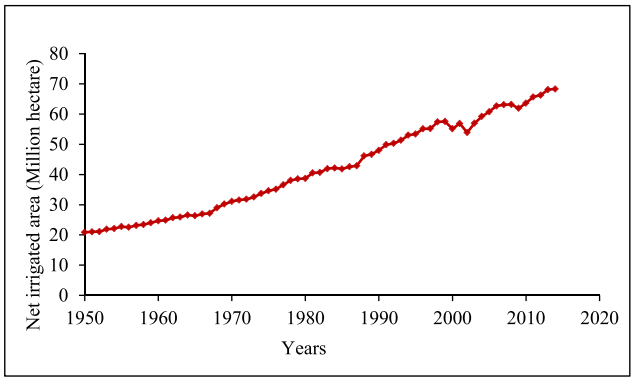


FIGURE 1 Expansion of net irrigated area in India. *Source:* FAOSTAT

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

America (including the Caribbean), yield exceeds 3 t/ha, while in South Asia, it is around 2 t/ha, and in sub-Saharan Africa, it just exceeds 1 t/ha (Wani et al., 2003). The dry subhumid and semi-arid regions of the world have experienced the lowest yields, with an average of 1–1.5 t/ha in South Asia for rainfed agriculture (Rockström et al., 2007; Wani et al., 2003). In India, productivity is particularly low, varying from 0.8 to 1.0 t/ha (Central Research Institute for Dryland Agriculture [CRIDA], 2011). There is a difference between growth rates of irrigated and rainfed agricultural systems of the country, which is reported as 1.1 t/ha in rainfed systems compared to 3.0 t/ha in irrigated systems (NRAA, 2020). In most of India's regions, irrigation water is provided during the non-monsoon season, while in some places it is supplied even during the monsoon in scarce and uncertain rainfall situations to get optimum crop yield. Besides high-quality seed and the right amount of fertilizer input, irrigation water management plays an indispensable role in enhancing crop productivity and has always been a subject of importance at the national level. At the farm level, the irrigation methods used in India may broadly be classified under the surface, sprinkler, and drip irrigation systems (Figure 2), of which surface irrigation by the flood and furrow methods is the most prevalent. Despite being a labour-intensive method with a lower water productivity of 30%–40%, surface irrigation covers more than 90% of the total irrigated area in India (Shah, 2011). With the facts understood well, several government policies, programmes, and schemes advocate scaling up modern irrigation facilities in India. However, farmers have not yet developed a sense of receptivity for modern irrigation techniques, for which they would have to abandon their age-old techniques which they received from their ancestors; this and the lack of communication between the policymakers and researchers have withheld the successful implementation of several such initiatives on the social and scientific front. The formulation of any scientifically and socially sustainable irrigation scheme and policy should be based on scientific inputs supported by pertinent research studies and have social acceptance. There are various documentations that highlight these problems and loopholes associated with irrigation project planning and development, implementation of irrigation schemes, creation and utilization of irrigation potential, development of command areas, and allocation and expenditure of funds at the regional and national levels (Shah et al., 2004; Kulkarni and Tyagi, 2012). The research gaps linked with irrigation at the farm level have yet to be addressed at the state and national levels, while most scientific work is being conducted at the laboratory/research farm scale under well-managed conditions with small-sized plots and controlled boundary conditions. But its validation at the field scale does not truly mimic the field conditions of farmers. This paper reviews and brings out the various research problems and researchable issues associated with irrigation on Indian farms with its meta-analysis. This paper also

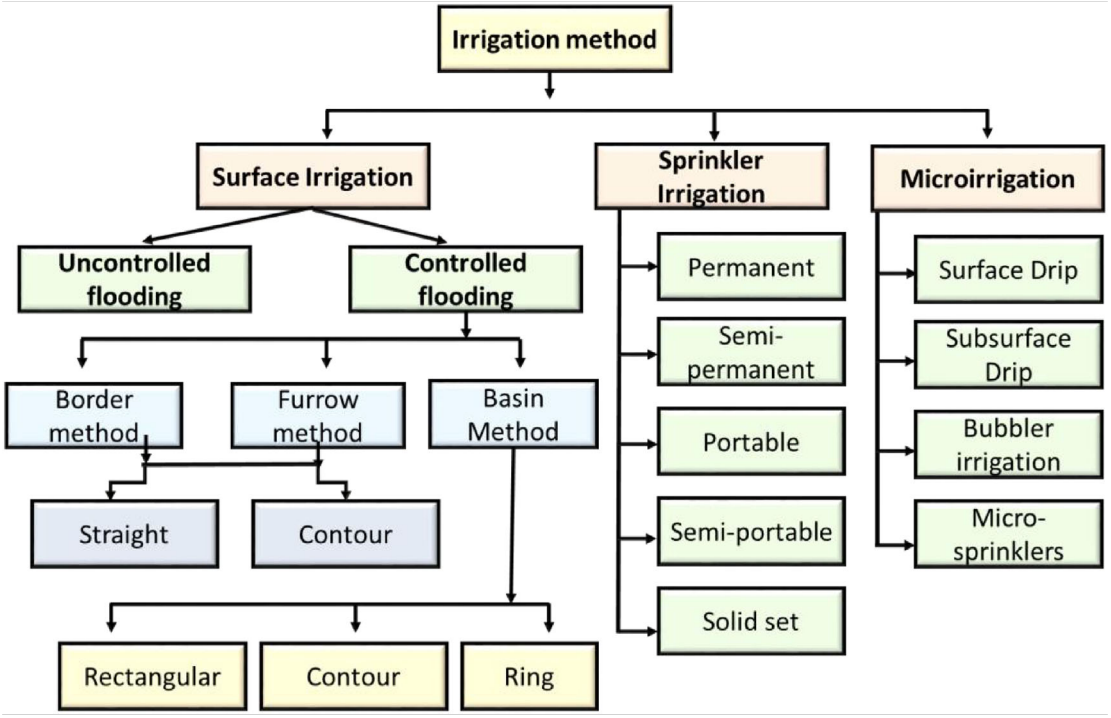


FIGURE 2 Common irrigation practices in India

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

assesses the current status of microirrigation (MI), evaluating the prospects of MI adoption in India. We start by elaborating the importance and overview of conventional surface irrigation methods in India, particularly focusing on methods of improving water productivity under surface irrigation systems. Further, we discuss the various researchable issues in surface irrigation at farmers' fields with the problems associated with canal and groundwater (GW) irrigation. Subsequently, the MI methods with their advantages and disadvantages have also been put forth. This section also covers the most common problems of farmers which hinder them from adopting these modern irrigation methods in the field. The last section overviews the government schemes and policies for developing irrigation and its management. Thus, this study analyses the current research gaps in addressing irrigation-related problems in the farmer's field vis-à-vis the researches to improve field water productivity.

2 | SURFACE IRRIGATION PRACTICES AND CHALLENGES

Conventional surface irrigation practices are employed for more than 90% of India's irrigated area, yet at present their obtainable application efficiency at the field is only about 35%–50% (Shah, 2011). Traditionally, crops are irrigated with surface methods of irrigation that is, furrow, border, and basin irrigation. The majority of the farmers continue using these methods due to the lower infrastructure and operational costs involved. However, since these methods have higher evaporation and percolation losses, the water use efficiency and crop water productivity are much lower. The various water-saving techniques such as alternate furrow irrigation (AFI), deficit irrigation (DI), and root zone drying irrigation, devised to improve water productivity and harvest quality, have been gradually, yet not widely, adopted over the years while complementing the conventional irrigation methods.

In India, irrigation infrastructure uses gravity flow canal distribution networks that distribute water from dams to low-lying command areas over a longer distance. Water lifting systems connected to pipe/open channel irrigation networks distribute water from stored or flowing water resources, such as open wells, underground tanks, ponds, traditional water storage structures, rivers, rainwater-harvesting systems, etc., for a relatively shorter distance.

The development of surface irrigation projects (diversion/storage types) defines the irrigation development work in the country, even though now the farmers' interest has shifted towards GW resources. The government participates in surface irrigation projects up to the watercourse, beyond which huge water conveyance losses are occurring all the way to farmers' fields, largely due to negligence in the management and operation of these systems. In the development and management of irrigation projects, various officials of different departments are involved; however, the poor collaboration and coordination among these departments impact the performance of irrigation projects. The inadequate attention to on-farm irrigation water management and utilization has been one of the main reasons for the poor performance of major and medium projects (Panigrahi and Panda, 2003). The most severe problem of Indian canal irrigation has been the rapid deterioration and degradation of existing network systems (Thakkar, 1999). This has also resulted in severe waterlogging and salinity problems in the canal command areas. The reported areas of 2.2 Mha under waterlogging and 3.5 Mha under salinity are in irrigation canal commands (Singh, 2005; Tyagi, 1997) and are further increasing due to inadequate attention towards drainage management. The estimated conveyance losses (approx. 45%) (Kulkarni et al., 2019) in these canal networks are also the major reason for the huge gap between actual and potential irrigation performance of surface irrigation projects in India. This gap is a force factor describing the urgent need to reform the government-managed irrigation systems (Central Water Commission [CWC], 2010).

The current agronomic practices with diversified cropping system (rice and non-rice crops) have led to the cultivation of different crops of different growth stages and different water needs under the same sluice/outlet. Thus, a common guideline of irrigation scheduling is practically not ideal for all the farmers under a command area, thereby necessitating the use of water metering mechanisms for customizing irrigation schedules that would increase water use efficiency. However, the lack of such control and measuring systems at the canal outlet heads and field outlets make farmers inept to measure the quantum of water irrigated at the field (Ray, 2011). This also leads to an irregular, unfair, and suboptimal distribution of water and a very low irrigation potential at the lower streams of the canal command area. Additionally, the formation and proper functioning of water users associations (WUA), ageing of irrigation and drainage infrastructure, and its operation and management (O&M) are the major organizational and public–private partnership challenges for sustainable water management in irrigated agriculture (Gany et al., 2019).

Scientific evidence-based planning, development, and management of command areas by primarily deriving the optimal outlet sizes according to the cultivated crops' water requirement is a prerequisite for service-orientated

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

sustainable management of water resources. Field adoption of scientifically validated irrigation scheduling methods with practically possible operating rules should increase the water use efficiency and overall system efficiency. The increased recurrence of meteorological droughts in India has reduced the availability of irrigation water in most canal command areas, thereby necessitating further drastic improvements over existing irrigation efficiencies while catering to the increasing areas brought under irrigation. Scientific evaluation to balance the overexploitation of water resources at the canal head with increasing coverage of the irrigated area at both upper and lower streams vis-à-vis socio-economic development of the command area are required across all existing irrigation canal networks. Such studies should provide a scientific basis for a fair distribution of limited water resources for enhancing water productivity, particularly during meteorological drought years.

In the canal command area, the scientific assessment of the water requirement of crops grown by the farmers and scheduling of irrigation should be based on soil–water–plant interactions (Duke and Crookston, 1987). The currently available irrigation scheduling technologies, though scientific, are expensive and relatively complex to comprehend by most farmers who have mid- to marginal-fragmented landholdings. Such farmers would readily adopt a simple, low-cost irrigation scheduling system or device with proper training for its O&M. The simplest approach to irrigation scheduling could be through using the evaporation pan (Allen et al., 1998) that can offer improved insight to irrigators about "when to water" based on estimated evapotranspiration values. But as this method excludes the effect of water table contribution and soil characteristics on water-holding capacity, it becomes difficult to define the exact irrigation intervals (Chattopadhyay and Hulme, 1997). Despite this, considering the limitations of on-demand irrigation in command area irrigation networks, the "how-much-to-irrigate" should be optimized to synchronize the "when-to-irrigate" approach for coinciding with canal water distribution cycles.

2.1 | Evaluation of surface irrigation efficiency

In terms of energy consumption, surface irrigation techniques are the best as they need either minor pumping to deliver water in the field or even no pumping if fields are properly graded as water advances under the gravity force. But, while considering the amount of water delivered in the field, a higher volume of water is generally applied to the field than the actual demand of the crop (El-Hazek, 2016). This is due to the time required for flows to advance across and along the field, and some extra time is needed to meet crop water demand at the tail end. This results in overwatering of the upper parts of the field with poor field water application efficiencies (Table 1). Some researchers have shown that surface irrigation efficiency can be as high as the efficiency of a well-managed sprinkler system, provided these systems are managed and operated efficiently (Bjorneberg, 2013; Levidow et al., 2014).

The major indicator of irrigation performance at the field is the intake characteristics of the soil during irrigation operation. Additionally, the performance of any irrigation system also varies greatly with field slope, channel shape, tractor traffic, soil moisture, crop type, and field history. However, the researchers have focused mainly on two variables for assessing the irrigation performance: inflow rate and duration (Walker, 1989; Ayars et al., 1999; Fangmeier et al., 1999).

The border/basin systems are the most common surface irrigation method for general crops and pastures. However, the furrow irrigation system is popular for row crops in India. The general performance measures of irrigation efficiencies are application efficiency and distribution efficiency. These indicators are applicable at the field scale. However, water losses occur in conveyance and distribution networks before reaching the field. Thus, the overall performance of irrigation efficiency should include conveyance, distribution, and storage efficiencies (Velasco-Muñoz et al., 2019). Also, the efficiency of irrigation water should be evaluated in a plant-physiological

TABLE 1 Irrigation efficiencies of different irrigation systems in India. *Source:* Jain et al. (2019); Kumar and Palanisami (2010); Sivanappan et al. (1994)

Irrigation efficiencies	Surface irrigation	Sprinkler irrigation	Drip irrigation
Conveyance efficiency (%)	40–50 (canal) 60–70 (well)	100	100
Application efficiency (%)	60–70	70–80	90
Soil moisture evaporation (%)	30–40	30–40	20–25
Overall efficiency (%)	30–35	50–60	80–90

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

sense, particularly assessing crop yield or economic return of an irrigated crop relative to the total amount of water transpired by the crop. The recent literature (Gleick et al., 2011; Gupta et al., 2017) is referring to this with the term *irrigation water productivity* instead of *water use efficiency*.

To evaluate the efficiency of border irrigation at the field, Narayana et al. (1980) mentioned that efficiencies calculated based on soil moisture variations give more realistic results than calculations based on opportunity time and infiltration. Later, the procedure overestimates the irrigation efficiencies due to empirical infiltration equations involved in it. They also recommended field outlet guidelines to get maximum efficiency at small farms with a discharge of 10 lps (litres per second): The border layout should be 50–70 m in length and 6–8 m in width.

Tyagi et al. (2012), during the study of the Bhakra Canal Project (India), reported a field channel discharge rate range from 20 lps to 100 lps and shared the views of Bishop and Long (1983) that with a discharge rate of more than 50–60 lps, conveyance losses are greater, while surface irrigation efficiency and uniformity is lower with a discharge rate of less than 20 lps. In the context of poor control of water flow, a typical problem in developing countries, Reddy (2013) reported the completion-of-advance to be more robust than the limiting length method of water application. Anwar et al. (2016) recommended that a depth of application of 50 mm leads to a low quarter distribution uniformity of 0.750 and application efficiency of 80%. Kaur et al. (2019) recommended a general relationship among flow rates, border lengths, border widths, and cut-off times based on field experiments and modelling to obtain optimum application efficiency. They reported that in light-soil fields with a length of 150 m, if the flow rate is known, the width (in meter) should be numerically half of the flow rate value (in lps), whereas, in medium soils, this value can be numerically equal to the flow rate. Also, at higher border lengths, the cut-off time (in minutes) can be numerically equal to the flow rate (in lps). They also reported that at higher border lengths, the cut-off time becomes insensitive to border widths and flow rates. Thus, the recommendations of research work highlighting relationships of inflow rate with the rate of advance, opportunity time, and cut-off time under different field layouts need to be adopted to improve field irrigation efficiency. All parameters affecting the field water application efficiency need to be optimized for different flow rates in a variety of conditions.

Similarly, instead of adopting a conventional single outlet water distribution system, farmers can adopt alternative methods to distribute water uniformly and efficiently in the field by introducing gated pipes and collapsible multi-outlet pipes (El-Motaleb et al., 2006; Karami and Samadi, 2005; Osman, 2002).

Some techniques for improving surface irrigation efficiencies are listed below.

2.2 | Irrigation water management for improving water productivity

Modern agronomic practices like AFI, raised bed sowing, mulching, laser land levelling; irrigation techniques such as partial root zone drying, regulated DI, alternate wetting and drying (AWD) method, and less water-consuming varieties have been validated to achieve sustainability in agricultural water use (Dhiman et al., 2015); hence these practices are promoted by the agriculture departments of the states and the central government through various programmes. Along with these practices, improved cultivation practices such as direct-seeded rice, conservation tillage, site-specific water and nutrient management, and variable-rate irrigation technologies are also being promoted in many parts of the country (Jat et al., 2020). Shifting from high water-guzzling crops such as lowland rice, cotton, and sugarcane to oilseeds, pulses, maize, forages, and agroforestry holds promise to increase water savings in irrigated agriculture. Based on the experience gained from a project conducted in the Indo-Gangetic Plain (IGP), using the participatory rural appraisal techniques, Rajput and Patel (2005) reported that through interventions such as laser levelling, renovation of watercourses, implementing optimal sowing dates for wheat, and demonstrations and training for farmers at least 30% (1.45 Mha·m) of irrigation water can be saved and an increase in wheat yield by at least 25% can be achieved in farmers' fields.

In Asia, researchers have shown that the average water need of the rice crop is about 1700 mm including soil preparation (FAO, 2003) for good growth, while in some literature it has been reported that traditional rice cultivation practice requires around 2000–2,400 mm of (Chauhan et al., 2012) water. The input of water in the paddy field is given for land saturation, puddling operation, maintaining the water layer, and to meet evaporation demand, transpiration, seepage, and percolation losses. The daily consumptive use of rice ranges from 6 to 10 mm depending upon the soil characteristics, duration of the variety, and the agroclimatic situation (FAO, 2003). The reported rice water use ranged from 1,570 mm in clay loam soil to 2,260 mm in sandy loam soil in India. However, these values in the IGP varies from 1,140 mm in Bihar to 1,560 mm in Haryana (Mandal et al., 2019). The total water requirement for

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

paddy ranged from 1,380 to 1,670 mm in the Palakkad district of humid tropical Kerala (Surendran et al., 2015).

As suggested by researchers, flood irrigation is not mandatory in rice cultivation. The other scientific cultivation methods available for rice cultivation being practised all over the world, namely, AWD, a system of rice intensification, and drip-irrigated dry-seeded rice (DSR), have the potential to save water and increase water productivity (Belder et al., 2004; Bouman and Tuong, 2001; Sharda et al., 2017). It has been reported by several authors that DSR technology consumes less water than puddled rice, without much effect on yield (Bhushan et al., 2007; Ravichandran et al., 2015; Yadav et al., 2010).

2.3 | Flow measurement and management

On-farm measurement of irrigation water flow and infiltration water uptake is essential to optimize surface irrigation performance parameters and to regulate the amount and duration of flow efficiently. This makes irrigators capable of better quantification of irrigation water used during an event (Replogle, 1997). However, nowhere on-farm water measurement is being practised at farm outlets in India. The line department and agriculture department had installed flumes at some places, but these are lying defunct due to negligence and improper care. The other reasons behind this problem are inadequate funds allocated for the "works" component under the O&M head in the surface irrigation schemes, lack of supervision and inspection by the line department, inefficient water rate recovery to meet O&M charges, negligence of WUA, and lack of farmers' awareness of care and maintenance of these structures (Planning Commission, 2002). In a surface irrigation system, the need to adopt real-time flow monitoring and automation in the irrigation system has been realized to cut deep percolation and run-off losses in the field. Some researchers have developed automatic flow meters, control meters, and automated gates with flow measuring units to accurately measure the water level and flow directed towards the field (Mishra and Raghavendra, 2014; Reddy et al., 2016; Sood et al., 2013). However, exhaustive trials at farmers' fields have not been mentioned, and robust commercial prototypes of these systems have not been produced. Thus, the adoption of automation in surface irrigation and flow metering at the farm channel is negligible, although it is known that automation in surface irrigation can facilitate various flow options such as constant flow, intermittent flow, surge flow, decreasing and dual flow, can better control cut-off timing, solve the problem of labour shortage, and save precious water resources (Evans and King, 2012). Under the project ClimaAdapt at the Water and Land Management Training and Research Institute (WALAMTARI), work for on-farm automatic measurement of water inflow and outflow from the field was done using low-cost ultrasonic sensors used with RBC (Replogle–Bos–Clemmens) flumes. These measurement units were installed at farmers' fields in the villages of Hyderabad district for better quantification of irrigation water use (Reddy et al., 2016). It is generally opined by several economists that the farmer would adopt modern technology only if the net return would be 30% higher than with the traditional technology. Therefore, economic and social research is also needed to understand the problems of developing countries to guide the engineers and scientists in creating automation solutions (Lowenberg-DeBoer et al., 2020).

2.4 | Land levelling

The use of laser technology for land levelling is a recent addition to Indian farms. The precise land levelling and well-maintained fields facilitate good irrigation scheduling practices, improve irrigation efficiencies, and reduce the wastage of water (Enciso et al., 2018). The recent progresses in laser land levelling practices enhance crop yield and water productivity and decrease labour requirement by reducing irrigation times because water distribution is even in a uniform field, which results in better seed germination, better crop growth, and higher crop yield (Jat et al., 2006). Approximately 25%–30% of irrigation water could be saved through uniform land levelling and grading without compromising crop yield (Asif et al., 2003; Bhatta and Sharma, 2009). It efficiently uses scarce water resources by eliminating the depression and uneven contours of the field, making them uniform (Naresh et al., 2014). A comparison of yield, water productivity, energy, and benefit–cost ratio (B:C) of laser land levelling with the traditional levelling method and an unlevelled field for wheat and rice crop as evaluated by Naresh et al. (2014) is presented in Table 2. The observations of irrigation time, amount, and frequency have been recorded at farmers' fields in several studies. However, the common practice of irrigators to cut off the water supply when the advance reaches the tail end leads to severe tail end run-off and deep percolation water losses and results in poor irrigation efficiency (Sharma and Singh, 2009). Therefore, though laser land levelling holds to be a proven technology, more studies for its thorough evaluation in terms of performance indicators, namely, application efficiency, storage efficiency, uniformity coefficient, and deep percolation need to be

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Parameters	Laser levelling		Traditional levelling		Unlevelled field	
	Rice	Wheat	Rice	Wheat	Rice	Wheat
Yield (t/ha)	5.7	4.5	5.4	4.2	4.3	3.8
WUE (kg m^{-3})	1.3	1.4	1.1	1.1	0.7	0.8
Energy (MJ ha^{-1})	4,770	2,500	5,660	2,890	6,960	3,650
B:C[a]	2.7	2.6	2.1	2.4	1.8	1.9

Abbreviations: B:C, benefit–cost ratio; WUE, water use efficiency.
[a]B:C is defined as the present value of the benefits expected from the project to the present value of the cost of the project, expressed in monetary terms.

TABLE 2 Comparison of precise land levelling with traditional levelling and unlevelled fields. *Source:* Naresh et al. (2014)

done along with devising a practical and efficient water metering method for use in levelled fields.

2.5 | Linking canal irrigation with pressurized irrigation systems

Conveyance of water through open channels and its application through surface irrigation in an existing canal system reduces the overall project efficiency to 30%–35% (Table 1). Therefore, improving the conveyance and application efficiencies in a surface irrigation system is a prerequisite for improving overall project efficiency and water productivity in the canal command area. Using pipes for conveyance and application eliminates all the associated losses, thereby increasing the project efficiency up to 89% (Wani et al., 2017). The success of a pressurized irrigation system in a canal command area relies upon the supply of water, switching from an on–off mode of water supply to a continuous water supply (Kumar et al., 2012). Thus, harvesting canal water into a reservoir in the field and reusing it through a pressurized irrigation system can improve water productivity in a canal-irrigated area, especially at the tail end. This rationale is supported by a study by Kumar et al. (2012), in which a drip-irrigated pomegranate orchard in a canal command area resulted in savings of 32% irrigation water and in 10.9% reduction in fruit cracking. The saved amount of water can be alternatively used for irrigating additional areas or can be diverted to meet the demands of the non-agricultural sector. Water productivity of a demand-synchronized system always ought to be more than of a supply-driven system. Thus, by introducing storage structures and drip/sprinkler systems, the canal supply-driven systems can be converted to demand-synchronized systems. Srivastava et al. (2010) observed that the irrigation efficiency can be increased up to 95% by replacing conventional irrigation methods with pressurized irrigation. Amarasinghe et al. (2012), in the canal command of the Indira Gandhi Nahar Pariyojana Project, Rajasthan, India, reported that on-farm water storage facilitated the use of a pressurized irrigation system with the flexibility of controlled timing and duration of irrigation; however, this required an irrigable area larger than 5 ha to be economically viable. Since the economic viability of such canal-cum-reservoir-based pressurized system depends on field condition, location, topography, and the annual cropping pattern, it is likely to be more feasible for areas characterized by coarse soil and/or rolling topography. Such areas otherwise have poor water productivity due to poor water-holding capacity and faster seepage of irrigation water down the slope under the canal water supply regime. Holistic simulation studies for optimizing the size of such water storage structures are required vis-à-vis economic constraints and environmental implications. Before upscaling this approach, the lessons from the case of missed goals of storage structures in "Jalyukt Shivar Abhiyan" of Maharashtra State have a much lower impact in drought-proofing the state (Bhadbhade et al., 2019) need to be observed scientifically. After establishing the site-specific merit of such systems through a scientific approach, the political willpower should take up promoting these in the canal command area by providing financial help for construction of stilling tanks, storage tanks, pumping houses, and laying of conveyance pipes from the canal outlets to the fields as an alternative to the construction of field channels. However, since the associated socio-economic and environmental implications are very vast, sufficient scientific evidence through pilot simulation and ground validity studies is needed first.

To design, develop, and operationalize such projects, the relevant factors related to the command area are cropping pattern, its water requirement, operating pressure of the system, desilting/filtration facilities, availability of power supply, relevant hardware and software systems, and capacity building through training of farmers. As of date, few pilot systems are working in the canal command areas of the Indian states of Rajasthan, Gujarat, Karnataka, and Himachal Pradesh. The recommendations are to take at least three crops annually with a drip-irrigated hybrid system to cut the payback period of the project. Therefore, from a techno-

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

economic perspective, further up-scaling of similar systems to other command areas would need either creating annual crop rotation plans and redesigning irrigation systems to match the requirements or installing a matching irrigation system to suit long-duration crops like sugarcane. One such attempt of up-scaling is seen in the state of Maharashtra that has issued guidelines to adopt MI in the canal command area for irrigating water-guzzling crops like sugarcane. In selected canal command areas, pilot projects are being conducted to improvise existing installations to run in hybrid mode as both surface and MI systems. The major issues to introduce MI in a canal command area are: (i) creating optimally sized water storage facilities at farm level assuring water availability for shorter irrigation intervals of 2–3 days; (ii) infrastructural changes for increasing reliability of electric power supply during irrigation cycles; (iii) financial support mechanism to cover the initial capital cost; and (iv) enforcing guidelines at the ground level to avoid suboptimal performance.

3 | GW IRRIGATION PRACTICES AND CHALLENGES

GW has emerged as a reliable and easily obtainable source of irrigation in India. It has complemented the green revolution at Indian farms that utilized water-intensive high-yielding crop varieties and rigorous cultivation practices (Shah, 1993). The use of GW resources for irrigation has seen a steep rise post independence (Figure 3). Much of this development happened in water-scarce basins and electrified places because of the inexpensive and unmetered electricity. The slow growth of the major irrigation projects of the country and poor maintenance of existing irrigation sources are also the reasons behind the steep rise in GW use (Moench, 2003). This source contributes about 44% of the total irrigation volume, but in return it provides nearly 60% of India's foodgrain production (Samra and Sharma, 2009; Shah, 2003).

According to the GW law of India, this subject is a private entity, and the right for utilization belongs to the owner of that piece of land (Jeet, 2005). Many researchers support the statement that crop productivity in areas irrigated with GW is one-third to one-half times more compared to areas irrigated by surface water resources (Dhawan, 2002; Gandhi and Bhamoriya, 2011; Siebert et al., 2010). This happens because of increased reliability of timely water availability, minimum conveyance losses, and the capacity of tube well irrigation compared to other irrigation systems (Shah, 1993). Most of the Indian farmers have little or no proper knowledge of GW hydrology and hydrogeology, but they certainly have some knowledge of GW withdrawal and its recharge underneath the earth surface (Deb Roy and Shah, 2002; Shah, 2011). The common pool nature and intangibility makes it hard to monitor, observe, and regulate GW resources, especially in developing countries.

States do not have any restrictions on GW use, installation of new pumping systems, and future GW usage conflicts. Until 2017, only half of the amount of water used was reported, when farmers started reporting water used in irrigated fields (V. Dhawan, 2017). The formulation of comprehensive strategies for scientific and sustainable GW resource management needs immediate attention to avert an aggravating GW crisis in India. The annual replenishable GW resource of the country is approximately 432 billion cubic meters (bcm), out of which 399 bcm is an available pool of GW for various developmental activities (Central Ground Water Board [CGWB], 2017). The irrigation sector is the main consumer (92%) of the available pool, with an estimated annual GW draft of 213 bcm. To achieve sustainability,

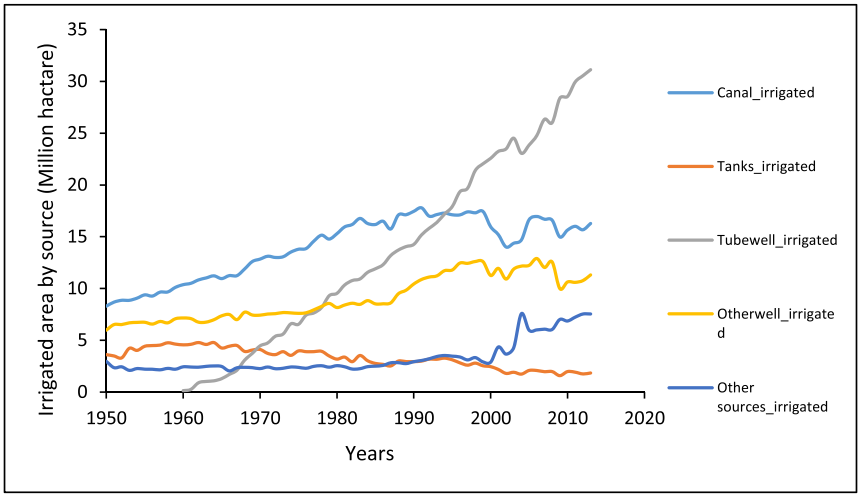


FIGURE 3 Statistics on net irrigated area by source of irrigation.
Source: Directorate of Economics & Statistics, Department of Agriculture and Co-operation, Ministry of Agriculture, Government of India

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

we need to "make sure that aquifers are at least half-full before a dry spell" (CGWB, 2017). At any place, GW development below 70% has been mentioned as "safe." To increase water productivity in an irrigated area, GW can safely be developed at least to the level of 60%–70% (Sharma and Singh, 2009).

In current practice, a frequent and large quantity of irrigation water applied in the poorly levelled fields without any prior knowledge of soil moisture and crop water requirement results in GW overdraft in most of the aquifers (Garduño et al., 2011). However, the combination of alternative irrigation practices, namely, drip irrigation, sprinkler irrigation, and furrow system with GW resources can greatly improve water productivity and crop yield. Since GW is largely extracted with pumping devices utilizing electricity, controlling the electric power supply for operating pumps can limit overexploitation. However, the off-grid solar pumping units, being popularized by the central and state governments, cannot be regulated with this analogy and would essentially require either connecting them with the internet of things (IoT) framework or bringing them on-grid/agricultural electricity feeder lines with load balancing and credit sharing mechanisms.

An estimate suggests 20% of the cropping intensity of India is prone to risk due to falling GW levels (Jain et al., 2021). Based on the data of NASA's Gravity Recovery and Climate Experiment (GRACE) from 2002 to 2016, scientists mapped the global trends in freshwater availability and reported that in northern India, the rate of GW depletion is 19.2 Gt/year, attributed primarily to water-intensive crops like rice and sugarcane (Rodell et al., 2018). The semi-arid region of northern India has the largest irrigated area (54%) among all the 34 regions studied worldwide. Though the rainfall of this region will increase by 12% by 2100, it has been quoted as "too little too late" to avert the crisis. Similarly, the eastern region of the country is suffering from a steep decline in GW because of the escalating rainfall pattern and excessive GW withdrawal for irrigation, and the trend will continue in the future (Rodell et al., 2018), which necessitates serious attention to policy reforms and their strict ground implementation to avert the anticipated crisis.

4 | MI AND SPRINKLER IRRIGATION PRACTICES AND CHALLENGES

MI methods are relatively new and offer benefits of enhanced water productivity and improved yields, with the added benefit of integrated fertigation functionality. In India, MI started penetrating its root in the 1980s (Palanisami and Raman, 2012). Typically, MI systems apply water at a high frequency and low rate either on the ground or underground. These systems include drip, bubbler, micro-sprinkler systems, etc., characterized by local (only root zone) application of irrigation water. However, the most prevailing MI system in India is drip irrigation. These systems operate under low pressure and require some form of pumping arrangements. In the drip method, a small quantity of water discharges through the outlet at a slow rate and emits directly to the crop root zone. These systems are water and energy efficient, which surely increases yields, input use efficiency, and economic returns of field crops (V. Sharma et al., 2019). The advantages of adopting MI and sprinkler irrigation over other traditional surface irrigation methods have been reported by several authors (Bassoi et al., 2003; Kumar and Palanisami, 2010; Pramanik et al., 2016). Various researchers have reported significant water savings using MI compared to the conventional irrigation methods for different crops in the diverse agroclimatic regions of the country (Ayars et al., 1999; Pandey et al., 2012; Madhava Chandran and Surendran, 2016). For instance, Deshmukh et al. (2010) reported 45% of water saving under drip irrigation over the conventional irrigation system in sugarcane cultivation.

The Task Force on Microirrigation (2004) estimated the potential area of MI and sprinkler irrigation in the country as 69.5 Mha (Table 3); however, the actual coverage estimated by the end of 2018 was approximately 11.4 Mha (GoI, 2020). Thus, the adoption of these irrigation methods on Indian farms is increasing at a slow pace of only 16.4% of the estimated potential area covered under MI and sprinkler irrigation. The promotion of MI is one of the major components of Pradhan Mantri Krishi Sinchai Yojana (PMKSY), a Government of India (GoI) scheme aimed at providing a comprehensive end-to-end solution for water management. Specifically, under the "Per Drop More Crop" component of this scheme, the GoI has made significant efforts to bring more area under MI and sprinkler irrigation for water-intensive row crops. Other related efforts are the convergence of MI and sprinkler irrigation with water-harvesting structures (tanks, ponds, wells, etc.) and the adoption of soil moisture conservation techniques and afforestation in overexploited and critical GW regions. For example, the Government of Haryana has made it compulsory to adopt a MI and sprinkler irrigation system with a water-harvesting structure in the 36 overexploited GW regions under the Mahatma Gandhi National Rural Employment Guarantee Scheme and the Integrated Watershed Development Programme.

The concept note prepared by NITI Ayog (Planning Commission, GoI) on "Micro Irrigation through Public–

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 Potential area of microirrigation in India. *Source:* Task Force on Microirrigation (2004)

Crops	Sprinkler area (Mha)	Drip area (Mha)	Total
Cereals	27.6	–	27.6
Pulses	7.6	–	7.6
Oilseeds	1.1	3.8	4.9
Cotton	1.8	7.0	8.8
Vegetables	2.4	3.6	6.0
Spices and condiments	1.0	1.4	2.4
Flowers, medicinal and aromatic plants	1.0	–	1.0
Sugarcane	–	4.3	4.3
Fruits	–	3.9	3.9
Coconut, oil palm, and plantation crops	–	3.0	3.0
Total	42.5	27.0	69.5

Private Partnership" highlighted the need for involvement of the private sector for O&M of installed MI networks for hassle-free operations. Similarly, the National Committee on the Use of Plastics in Agriculture (NCPA), GoI, has recommended the use of high-tech horticulture and precision farming (greenhouse, vertical farming, etc.) for selected crops to achieving greater productivity of horticulture crops (Suresh and Samuel, 2020).

4.1 | Yield and productivity under drip irrigation

Drip irrigation offers substantial savings of electricity and water with improved yield compared with conventional surface irrigation (Ashoka et al., 2015; Madhava Chandran and Surendran, 2016). Experimental field studies conducted on different crops reported 30%–57% water savings (Madramootoo and Morrison, 2013) and a 20%–90% increase in yield (Sivanappan et al., 1994) using drip irrigation over surface irrigation. However, the performance of irrigation will be lower in farmers' fields due to poor controlling factors.

A survey of 200 farmers from Maharashtra state reported water savings in sugarcane (44%), grapes (37%), and banana (29%) for drip irrigation over conventional irrigation methods (Narayanamoorthy, 2007). It was also reported that a pump-irrigated DI sugarcane field consumes 1.28 HP-hr of water (to measure consumption of water at a field, the unit *HP-hours of water* is computed by multiplying the HP of the pumpset with the hours of water usage) for producing 100 kg of sugarcane against 2.83 HP-hr of water in surface irrigation methods.

There are some studies in the literature that have evaluated the DI system in terms of different parameters, namely, water consumption (Gupta et al., 2019), electricity consumption (Madhava Chandran and Surendran, 2016), and economics (Srivastava and Upadhayaya, 1998); however, there are limited studies that have analysed the impact of the DI method on electricity saving and economic viability using field survey data of farmers' fields (Narayanamoorthy, 2007). Sivanappan et al. (1994) have provided B:C for different crops under DI and suggested that DI is economically viable, even if the cost of water saving is omitted from the calculations. There is always room for reducing the capital cost of a MI system to suit small and medium farms by incorporating some modifications in farming practices, such as adopting paired-row crop planting.

Subsurface drip irrigation (SDI) has emerged as an advanced technique over surface drip irrigation. It significantly reduces direct evaporation, run-off, and deep percolation losses (Hanson and May, 2007; Safi et al., 2007). SDI may cover an area of about 1.5 times the area covered under surface drip irrigation with the same amount of water. SDI has emerged as a very good irrigation technique in terms of reducing the total water requirement, improving the fertilizer application efficiency, and reducing the weed population (Souza and Bizari, 2018). Its use has resulted in increased yield and water savings compared to other irrigation systems. SDI has more soil water stability in the middle and deep soil layers, causing improvement in water productivity, in comparison to surface drip irrigation (Yang et al., 2019). Mahesh et al. (2016) reported that in sugarcane cultivation, SDI and surface drip irrigation can save 31% and 23% of irrigation water, respectively, compared to the surface irrigation methods. However, SDI shows some lacunae, such as low germination in hard soil due to poor capillary movement, salinity deposition around the crop root zone, emitter clogging, and uneven water

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

distribution (Kaushal et al., 2012). SDI does not always guarantee high efficiency and good yield since it requires an accurate design and a skilled operator (Dlamini, 2005).

In the Indian situation, the MI system has been mainly adopted for orchards; vegetables; nurseries; plantation crops such as mango, guava, papaya, grapes, orange, pomegranate, rubber, tea, and coffee; in-field crops such as sugarcane, cotton, and groundnut; and protected cultivation (Raman, 2010).

4.2 | Yield and productivity under sprinkler irrigation

The pressurized sprinkler irrigation systems provide more uniform water distribution in the field than the surface irrigation method. In India, the sprinkler irrigation system is mainly adopted for field crops (e.g., wheat, groundnut, sorghum, etc.), row crops, and trees for irrigating water under or over the canopy (Shiva Shankar et al., 2015). However, delicate crops such as flowers and vegetables are generally not irrigated with sprinkler irrigation (M. Kumar et al., 2009). Several studies have compared sprinkler irrigation with conventional surface irrigation methods for different crops and reported an increase in yield (5%–50%) and water savings (10%–40%) and thus higher water productivity (10%–50%) with sprinkler irrigation (Acharyna et al., 1993; Dhawan, 2002; Ramanjaneyulu et al., 2013; Shiva Shankar et al., 2015). Sharma et al. (2018) reported that sprinkler irrigation provides 12.5% irrigation water saving, 16.2% increase in yield, and 30.7% increase in water productivity over the border irrigation method for irrigated wheat. The sprinkler irrigation system is best suited for soils with high infiltration rate, and the application rate of sprinklers should be set to match the basic infiltration rate of soil (NCPA, 1990).

Sprinkler irrigation is also used during the crop season when there is a shortage of irrigation water, just before the onset of the monsoon for crop sowing or in case of a long dry spell during monsoon.

The on-field performance and uniformity coefficient of the sprinkler irrigation system can deteriorate compared to the potential due to several factors beyond the operational control of farmers, such as fluctuations in wind speed, wind direction, the hydraulic pressure of the system, etc. (Figure 4). The experimental and research data to overcome these problems can require complex processing and analysis to improve system design and field performance (Madramootoo and Morrison, 2013). Similarly, the lack of knowledge of the farmers regarding operating conditions, the relationship between operating pressure and flow, and essential settings to change the mode of the system operation reduce on-field performance. Pumping system type, size of sprinklers, and operating pressure should fulfil the crop water requirement and enable the uniform application of water (Indian National Committee on Irrigation and Drainage [INCID], 1998).

4.3 | Automation in MI

The recent developments in MI integrate electronics, software, and different sensors capturing real-time field data for automating its operation on the field. These advanced techniques reduce energy consumption and

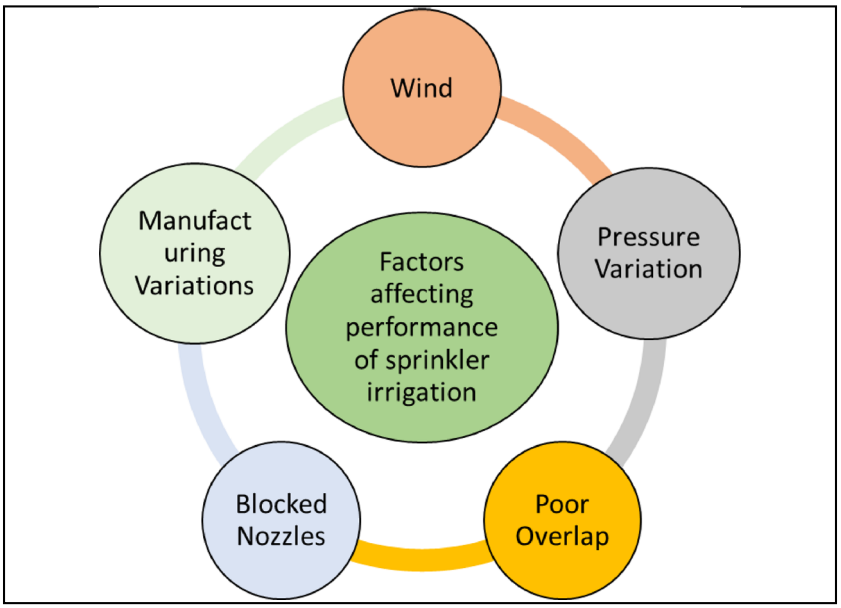


FIGURE 4 Factors influencing uniformity and performance of sprinkler irrigation systems

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

costs of operation and provide data security, scalability, remote monitoring, and telemetry work. The addition of soil moisture monitoring sensors and soil temperature recorders can provide more accurate decisions on irrigation scheduling in a microirrigated field with some extra cost of added equipment. Earlier researchers have performed automation in the fields using MI based on sensors and integrated programming (Abraham et al., 2000). Luthra et al. (1997) designed an efficient valve control system for drip automation and fertigation. Further, Joshi et al. (1999) patented an automatic controller. Joshi et al. (2002) and Reddy et al. (2016) did research work for automation in drip irrigation. Since the higher cost of the sensors has been the major limiting factor for its field adoption, research for cost-effective solutions should lead to more field implementation. With the advancement in electronics and instrumentation, several researchers have developed an automatic MI system for irrigation scheduling and fertigation using the IoT, machine learning, Raspberry Pi, Arduino, Wi-Fi, and renewable energy sources (Agrawal et al., 2014; Agrawal & Singhal, 2015; Sharma et al., 2019; Sudharshan et al., 2019; Vij et al., 2020). Several private industries have been providing services for automatic MI systems in India, most notably Jain Irrigation Private Ltd. and Netafim Irrigation Ltd.

Bombale et al. (2011) evaluated the automatic drip irrigation system at farmers' fields and reported that voltage fluctuation, wiring damage, valve breakage, and lack of repairing services are the main problems faced by farmers in the field. Long-term field evaluation of these and similar automation devices have not been reported in the farmer's field, and its reliability for long-term functioning is uncertain. Thus, the majority of farmers prefer the conventional method in MI. The soil properties, chemical properties of irrigation water, field conditions, and cost economics should be assessed properly before selecting any sensing probes. Since the use of these sensors requires some prior knowledge about instrumentation and its operations, the irrigators should be given fundamental training and consultations for easy operation and maintenance. Also, the increased pace of innovations in sensing solutions using newer hardware and software tools may phase out older designs and support by companies and therefore may allow only limited years of on-field functioning of the automated solutions developed by researchers and companies.

4.4 | System maintenance and equipment quality

The quality and certified standards (SI standards) of the irrigation equipment are important considerations for selection and installation in the field, especially when fertigation is to be done with the existing system. MI methods ensure high uniformity in water distribution and provide accurate irrigation scheduling until the system performs well. Thus, it becomes imperative to select the irrigation equipment that accurately suits the pedological, agronomical, and soil conditions of the field. The complexity of MI systems lies in their design and not in their operation (Kumar et al., 2009). For smooth operation of the system, basic skills of the farmers, which may include maintenance, filter cleaning and changing, controlling emitter flow, and operating the flushing system, are required to enhance its life span.

The major technical issue with MI operation in the field occurs due to clogging of emitters and lateral pipes with suspended sand, silt, and or algae entering with water and from the chemical precipitation of Mg and Ca, when irrigated with hard water. Solving these at the farmer's end, require availability and accessibility of good technical support and assistance, and this remains a scaling challenge in a hugely populated country like India.

4.5 | Challenges in adoption of MI

There are three pillars for the successful adoption of MI technology on Indian farms: (a) economically viable and technically sound technology; (b) awareness towards the technology among the user group; and (c) selected technology should be easily accessible by users (Namara et al., 2005). The major challenges in the adoption of MI faced by the farmers of the country are the high initial investment, delay in the subsidy distribution; uncertain government guidelines, difficult financing mechanism; poor follow-up services by agencies, as well as delays in the loan sanctioning process by banks (Namara et al., 2007). Out of these challenges, the high initial cost of investment secures the first rank for non-adoption of MI in the field by Indian farmers (Madhava Chandran and Surendran, 2016; Polak et al., 1997; Suresh et al., 2019). The central and state governments are working closely to tackle such problems. The government has not only announced a large subsidy for MI but also is closely working with MI industries to control the unit costs of manufacturing. Also, the local manufacturers are being encouraged to supply the main and spare parts of the MI systems to the farmers.

In India, a sizeable group of farmers, even in the present era of information technology, are not acquainted with the benefits of adopting modern tools and techniques. Those who know about it, find the system too difficult to manage and operate. There is a lack of trained hands too. The extension agencies can widen the

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

dissemination of know-how about the advances in MI through consistent media campaigns and through online training videos which can now easily be reached by farmers with affordable last-mile internet connectivity made possible through the telecom revolution in India during the last half decade. This should also be supported with hands-on training for the use and management of MI at Krishi Vigyan Kendra (KVK) covering all the districts of India. However, this would also need proven sound and replicable protocols for crop-specific irrigation schedules for the different agroclimatic zones of the country. Such protocols should be compatible with existing or improved management practices including irrigation, fertilization, pest management, and harvesting followed by farmers or recommended for the area.

Many institutions have been working over the last few decades to improve the infrastructure of irrigation systems and management and utilization of irrigation water in India. Many WUA have been set up across the states of the county for inspection and management of irrigation systems. Realizing the need of farmers in irrigation system management, the participatory irrigation management approach is practised in most of the states. The nation-wide target for doubling farmer income (DFI) in the DFI Committee Report (2018) recommends achieving it by the judicious and scientific application of water. To support DFI, the PMKSY scheme of the GoI will bring more areas under the MI system as the objective of one of its schemes ("per drop more crop"). Under this scheme, a dedicated microirrigation fund with an initial corpus of Rs. 5,000 crores (Rs. 2,000 crores for 2018–2019 and Rs. 3,000 crores for 2019–2020) has been released to provide resources to the states for expanding coverage under MI.

5 | IMPLEMENTATION OF RESEARCH FINDINGS

Despite the good number of studies on on-farm water management and their adoption at the farm scale by the farmers, there are still gaps to be addressed. The on-farm adoption rate of better water management practices is still slow and limited by gaps in research and extension arenas. For bridging the gaps between research and its practice at the farm level, the following need to be prioritized:

i. Revival of old indigenous water storage/use technologies and their integration with the latest technologies;
ii. Development of decision support systems for judicious utilization of rain, surface, and GW resources;
iii. Development of cost-effective water management technologies/practices suitable for small landholders;
iv. Ways to improve water productivity and implementation of interventions specific to different agroclimatic zones;
v. Capacity building and training of water users and water suppliers; and
vi. Transfer of socially acceptable and economically viable sustainable water management technologies to farmers' fields and their effective implementation and up-scaling using a participatory approach.

6 | CONCLUSIONS

This paper reviewed the challenges of water management in farmers' fields in India and explores possibilities for higher irrigation use efficiency. A significant amount of water is lost in the conveyance and distribution channels of the surface irrigation systems, resulting in poor irrigation efficiency in the field. Thus, systematic maintenance of these channels, proper irrigation scheduling, precise land levelling, right cultivation methods, effective drainage, and efficient management of water by the farmers are the main focus areas for improving water productivity in the surface irrigation system. This review puts forth the potential of water-saving technologies such as, precision levelling, and automation in irrigation systems as reported by researchers. However, scaled studies of farmer's field conditions to support the findings of researchers are meagre. At the watershed level, the overall reduction in water consumption in irrigation is probably achieved when combinations of different water-saving technologies are used by the irrigators. The industries and the scientific community work together in a collaborative mode to fulfil the needs and demands of the present situation. As a result, various technologies related to the irrigation sector have been developed, which include sensors for measurement of soil moisture, weather, and plant parameters and various data recording units that log/transmit data to irrigators continuously or intermittently for manual or automated irrigation scheduling using flow controllers and regulators. However, the adoption of such technologies at the farmer's level is low or negligible.

Though per se the modern technologies are available, poor access to these creates a serious hurdle in their adoption. The inadequate access to public irrigation facilities, timely electric power supply, supportive mechanization solutions, the limited reach of the facilities of command area development work up to the last-mile small landholdings, and poor on-farm water

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

management are the main reasons behind the non-adoption of modern irrigation technologies on the farms of the country. Furthermore, a change of irrigation systems must start at the policy level. The concept of farm water entitlement and flow-metering standards of irrigation water should be introduced in law and rules. Reforming the irrigation institutions and agencies for their functioning in a transparent manner, where farmer participation is given priority, would enhance the efficiency and productivity of an irrigation system. The failure to recognize the need for technological improvements and the required degree of reformation in any irrigation project would create a large gap between creation and utilization of irrigation potential. Besides, capacity building and human skill development of the basic scientific know-how about irrigation systems need to be disseminated to the farmers and irrigators at a larger scale. A basic understanding of surface water and GW hydrology is critical for irrigators because it allows better management of available water resources as well as solving water recharging problems. Thus, in the present situation, adoption of modern irrigation practices at the farm level, right advisory on irrigation, automation in the field, as well as training and awareness programmes for the farmer community are solutions to address the prevailing issues. This would not only save scarce water resources but also greatly address the prevalent labour shortage, improve work efficiency, and increase farmers' income. Thus, the outcome of this review analysis would provide substantial insights to irrigation researchers and policymakers during new irrigation project and government policy formulations for adoption of water-efficient irrigation technologies by farmers of India.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study

ORCID

Ajita Gupta https://orcid.org/0000-0002-7340-663X
Mukesh Kumar https://orcid.org/0000-0002-4416-6198
Chetankumar Prakash Sawant https://orcid.org/0000-0002-1006-7415

REFERENCES

Abraham, N., Hema, P.S., Saritha, E.K. & Subramannian, S. (2000) Irrigation automation based on soil electrical conductivity and leaf temperature. *Agricultural Water Management*, 45(2), 145–157.

Acharyna, M.S., Singh, J., Gupta, A.P. & Singh, A.K. (1993) *Economic analysis of sprinkler irrigation in Southern Rajasthan. in CBIP (1993), Sprinkler and drip irrigation systems*. New Delhi: Central Board of Irrigation and Power, pp. 74–76.

Agrawal, N. & Singhal, S. (2015) Smart drip irrigation system using Raspberry Pi and Arduino. *International Conference on Computing, Communication and Automation, ICCCA*, 2015, 928–932. https://doi.org/10.1109/CCAA.2015.7148526

Agrawal, S., Pandey, S., Kumar, S. & Chaudhery, K. (2014) Automatic irrigation system. *International Journal of Advance Research and Innovation*, 2(1), 92–94.

Allen, R.G., Pereira, L.S., Raes, D. & Smith, M. (1998) Crop evapotranspiration (guidelines for computing crop water requirements). FAO Irrigation and Drainage Paper No. 56. 13(3), 110–115. https://doi.org/10.1016/S0141-1187(05)80058-6

Amarasinghe, U.A., Palanisami, K. & Singh, O.P. (2012) Improving canal irrigation performance with on-farm water storage: evidence from the Indira Gandhi Nahar Pariyojana project in India. *Irrigation and Drainage*, 61(4), 427–435. https://doi.org/10.1002/ird.689

Anwar, A.A., Ahmad, W., Bhatti, M.T. & Ul Haq, Z. (2016) The potential of precision surface irrigation in the Indus Basin Irrigation System. *Irrigation Science*, 34(5), 379–396. https://doi.org/10.1007/s00271-016-0509-5

Ashoka, P., Kadasiddappa, M.M. & Sanjey, M.T. (2015) Enhancing water productivity through micro-irrigation technologies in Indian agriculture. *Annals of Plant and Soil Research*, 17(Special issue), 601–605.

Asif, M., Ahmed, M., Gafoor, A. & Aslam, Z. (2003) Wheat productivity, land and water use efficiency by traditional and laser land-leveling techniques. *Online Journal of Biological Science*, 3(2), 141–146.

Ayars, J.E., Hutmacher, R.B., Schoneman, R.A., Soppe, R.W.O., Vail, S.S. & Dale, F. (1999) Realizing the Potential of Integrated Irrigation and Drainage Water Management for Meeting Crop Water Requirements in Semi-Arid and Arid Areas. *Irrigation and Drainage Systems*, 13(4), 321–347. https://doi.org/10.1023/a:1006390421589

Bassoi, L.H., Hopmans, J.W., de Jorge, L.A., de Alencar, C.M. & Silva, J.A.M. (2003) Grapevine root distribution in drip and microsprinkler irrigation. *Scientia Agricola*, 60(2), 377–387. https://doi.org/10.1590/s0103-90162003000200024

Belder, P., Bouman, B.A.M., Cabangon, R., Guoan, L., Quilang, E.J.P., Li, Y. *et al.* (2004) Effect of water saving irrigation on rice yield and water use in typical lowland conditions in Asia. *Agricultural Water Management*, 65(3), 193–210. https://doi.org/10.1016/j.agwat.2003.09.002

Bhadbhade, N., Bhagat, S., Joy, K.J., Samuel, A., Lohakare, K. & Adagale, R. (2019) Can Jalyukt Shivar Abhiyan prevent drought in Maharashtra? *Economic & Political Weekly*, 25, 12–14.

Bhatta, R. & Sharma, M. (2009) Laser leveller for precision land levelling for judicious use of water in Punjab. In: *Extension Bulletin, Krishi Vigyan Kendra, Kapurthala*. Ludhiana: Punjab Agricultural University.

Bhushan, L., Ladha, J.K., Gupta, R.K., Singh, S., Tirol-Padre, A., Saharawat, Y.S. *et al.* (2007) Saving of water and labor in a rice–wheat system with no-tillage and direct seeding technologies. *Agronomy Journal*, 99(5), 1288–1296. https://doi.org/10.2134/agronj2006.0227

Bishop, A.A. & Long, A.K. (1983) Irrigation water delivery for equity between users. *Journal of Irrigation and Drainage Engineering*, 109(4), 349–356. https://doi.org/10.1061/(ASCE)0733-9437(1983)109:4(349)

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Bjorneberg, D.L. (2013) Irrigation|methods. In: *Reference Module in Earth Systems and Environmental Sciences*. USDA Agricultural Research Service, Kimberly, ID, USA: Elsevier Inc. https://doi.org/10.1016/b978-0-12-409548-9.05195-2

Bombale, V.T., Popale, P.G., Magar, A.P. & Chavan, M.L. (2011) Surve of automation irrigation systems in Maharashtra. *Engineering and Technology in India*, 2(1&2), 58–60.

Bouman, B.A.M. & Tuong, T.P. (2001) Field water management to save water and increase its productivity in irrigated lowland rice. *Agricultural Water Management*, 49(1), 11–30. https://doi.org/10.1016/S0378-3774(00)00128-1

Central Ground Water Board (CGWB). (2017) *Dynamic ground water resources of India (as on 31st March 2013)*. India: Ministry of Water Resources, River Development and Ganga Rejuvenation, Government of India. Faridabad.

Central Research Institute for Dryland Agriculture (CRIDA). (2011) *Vision 2030*. Hyderabad, India: Santoshnagar, p. 31.

Central Water Commission (CWC). (2010) Financial aspects of irrigation projects in India.

Chand, R., Srivastava, S.K. & Singh, J. (2017) Changing Structure of Rural Economy of India- Implications for Employment and Growth. Delhi, India: National Institution for Transforming India (NITI) Aayog, 1–26.

Chattopadhyay, N. & Hulme, M. (1997) Evaporation and potential evapotranspiration in India under conditions of recent and future climate change. *Agricultural and Forest Meteorology*, 87(1), 55–73. https://doi.org/10.1016/S0168-1923(97)00006-3

Chauhan, B.S., Mahajan, G., Sardana, V., Timsina, J. & Jat, M.L. (2012) Chapter six - Productivity and sustainability of the rice–wheat cropping system in the Indo-Gangetic Plains of the Indian subcontinent: problems, opportunities, and strategies. *Advances in Agronomy*, 117, 315–369. https://doi.org/10.1016/B978-0-12-394278-4.00006-4

Das, B.P. (2003) The Use of Irrigation Systems Water Resources of India and Irrigation. Fisheries in Irrigation Systems of Arid Asia. FAO Fisheries Technical Paper. No. 430. Rome, FAO. 47–70.

Deb Roy, A. & Shah, T. (2002) The socio-ecology of groundwater in India. *Water Policy Briefing*, 4, 1–6.

Deshmukh, A.S., Shinde, P.P., Katake, S.S., Phonde, D.B., Mali, V. S. & Imas, P. (2010) Effect of various levels of potash application through drip irrigation on yield and quality of sugarcane. *International Potash Institute*, 24(September), 1–6.

Dhawan, B.D. (2002) *Technological change in Indian irrigated agriculture: a study of water saving methods*. New Delhi, India: Commonwealth Publishers.

Dhawan, V. (2017) Water and agriculture in India. Background Paper for the South Asia Expert Panel during the Global Forum for Food and Agriculture, 28. https://www.oav.de/fileadmin/user_upload/5_Publikationen/5_Studien/170118_Study_Water_Agriculture_India.pdf

Dhiman, J., Dhiman, J.S., Aggarwal, R. & Dhiman, M. (2015) Water management for sustainability of irrigated agriculture: an Indian perspective. *Indian Journal of Economics and Development*, 11(3), 601–622. https://doi.org/10.5958/2322-0430.2015.00069.4

Dlamini, M. (2005) Experience with drip irrigation on smallholder sugarcane irrigation schemes in Swaziland. *Proc. S. Afr. Sugar Technol. Assoc*, 79, 463–472.

Duke, H.R. & Crookston, M.A. (1987) *Scheduling irrigations: a guide for improved irrigation water management through proper timing and amount of water application*. Colorado, United States: Soil Conservation Service; Agricultural Research Service: Cooperative Extension (CSU).

El-Hazek, A. (2016) Challenges for optimum design of surface irrigation systems. *Journal of Scientific Research and Reports*, 11(6), 1–9. https://doi.org/10.9734/jsrr/2016/27504

El-Motaleb, I. A. A., El-Saadaw, M. A., Hindy, F. I. & Mattar, M. A. (2006) Development of perforated pipes to improve surface irrigation performance - new trends in agricultural engineering, 1085–1106. The 14th. Annual Conference of the Misr. Society of Ag. Eng., 22 Nov. 2006.

Enciso, J., Jung, J., Chang, A., Chavez, J.C., Yeom, J., Landivar, J. & Cavazos, G. (2018) Assessing land leveling needs and performance with unmanned aerial system. *Journal of Applied Remote Sensing*, 12(1), 1–8. https://doi.org/10.1117/1.JRS.12.016001

Evans, R.G. & King, B.A. (2012) Site-specific sprinkler irrigation in a water-limited future. *Transactions of the ASABE*, 55(2), 493–504. https://doi.org/10.13031/2013.41382

Fangmeier, D.J., Clemmens, A., El-Ansary, M.S., Strelkoff, T.E. & Osman, H. (1999) Influence of land leveling precision on level-basin advance and performance. *Transactions of ASAE*, 42(4), 1019–1026. https://doi.org/10.13031/2013.13249

Food and Agriculture Organization of the United Nations (FAO). (2003) *Rice irrigation in the Near East: current situation and prospects for improvement*. Cairo, Egypt: FAO Regional Office for the Near East.

Food and Agriculture Organization of the United Nations (FAO). (2015) *Towards a water and food secure future. Critical perspectives for policy-makers*. Rome: FAO and Water World Council. http://www.fao.org/3/a-i4560e.pdf

Food and Agriculture Organization of the United Nations (FAO). (2017) Water for sustainable food and agriculture. A Report produced for the G20 Presidency of Germany. Rome. http://www.fao.org/3/i7959e/i7959e.pdf

Gandhi, V.P. & Bhamoriya, V. (2011) Groundwater irrigation in India: growth, challenges and risks. *India Infrastructure Report*, 2011, 90–117. http://www.idfc.com/foundation/policy/india_infrastructure_report.htm

Gany, A.H.A., Sharma, P. & Singh, S. (2019) Global review of institutional reforms in the irrigation sector for sustainable agricultural water management, including water users' associations. *Irrigation and Drainage*, 68(1), 84–97. https://doi.org/10.1002/ird.2305

Garduño, H., Romani, S., Sengupta, B., Tuinhof, A. & Devis, R. (2011) India groundwater governance case study (Issue June). www.worldbank.org/water

Gleick, P.H., Christian-Smith, J. & Cooley, H. (2011) Water-use efficiency and productivity: Rethinking the basin approach. *Water International*, 36(7), 784–798. https://doi.org/10.1080/02508060.2011.631873

Government of India (GoI). (2018) Pocket book of agricultural statistics 2017. https://eands.dacnet.nic.in/PDF/Agricultural_Statistics_2017.pdf

Government of India (GoI). (2020) Pocket book of agricultural statistics 2019. https://eands.dacnet.nic.in/PDF/PocketBook2019.pdf

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Gupta, A., Rao, K.V.R., Singh, S., Soni, K. & Sawant, C. (2019) Water productivity and yield of baby corn (*Zea mays L*) as influenced by irrigation levels under subsurface drip irrigation. *International Journal of Chemical Studies*, 7(5), 128–135.

Gupta, A., Sarangi, A. & Singh, D.K. (2017) Estimation of crop coefficients and water productivity of mustard (*Brassica juncea*) under semi-arid conditions. *Current Science*, 113(2), 1–8.

Hanson, B. & May, D. (2007) The effect of drip line placement on yield and quality of drip-irrigated processing tomatoes. *Irrigation and Drainage Systems*, 21(2), 109–118. https://doi.org/10.1007/s10795-007-9023-5

Indian National Committee on Irrigation and Drainage (INCID). (1998) Sprinkler irrigation in India, New Delhi, India.

Jain, M., Fishman, R., Mondal, P., Galford, G.L., Bhattarai, N., Naeem, S. *et al*. (2021) Groundwater depletion will reduce cropping intensity in India. *Science Advances*, 7, 1–9.

Jain, R., Kishore, P. & Singh, D.K. (2019) Irrigation in India: status, challenges and options. *Journal of Soil and Water Conservation*, 18(4), 1–10. https://doi.org/10.5958/2455-7145.2019.00050.x

Jat, M.L., Chandna, P., Gupta, R., Sharma, S.K. & Gill, M.A. (2006) Laser land leveling: a precursor technology for resource conservation. *Rice-Wheat Consortium (Technical Bulletin Series 7)*, 7, 48.

Jat, M.L., Jat, H.S., Agarwal, T., Bijarniya, D., Kakraliya, S.K., Choudhary, K.M. *et al*. (2020) *A compendium of key climate smart agriculture practices in intensive cereal based systems of South Asia. P.42*. New Delhi, India: International Maize and Wheat Improvement Center (CIMMYT).

Jeet, I. (2005) *Groundwater resources of India: occurrence, utilisation, and management*. New Delhi, India: Mittal Publications. https://books.google.co.in/books?id=dIMeAQAAIAAJ

Joshi, A., Tiwari, K.N. & Banerjee, S. (1999) Automated irrigation controller (Patent No. # 6/Cal/99 dated 04-1-1999).

Joshi, A., Tiwari, K.N. & Banerjee, S. (2002) Low cost computer controlled automated irrigation system (Patent No. # 307/Cal/2002 dated 15-5-2002).

Karami, V. & Samadi, B.R. (2005) Improved surface irrigation methods by using gated pipes. Technical workshop automated surface irrigation, 209-221. Tehran, Iran. *(in Farsi)*

Kaur, S., Raheja, A. & Aggarwal, R. (2019) Performance evaluation and optimization studies of border irrigation system for wheat in the Indian Punjab. *Water SA*, 45(1 January), 41–47. https://doi.org/10.4314/wsa.v45i1.05

Kaushal, A., Patole, R. & Singh, K.G. (2012) Drip irrigation in sugarcane: A review. *Agricultural Reviews*, 33(211–219), 2012.

Kulkarni, A.A., Nagarajan, R. & Rabelo, L.C. (2019) Hydrospatial modelling and simulations for assessing the irrigation canal conveyance losses. *Modelling and Simulation in Engineering*, 4, 1–7. https://doi.org/10.1155/2019/1803748

Kulkarni, S.A. & Tyagi, A.C. (2012) Participatory Irrigation Management: Understanding the Role of Cooperative Culture. In International Commission on Irrigation and Drainage, 1–8.

Kumar, D.S. & Palanisami, K. (2010) Impact of drip irrigation on farming system: evidence from Southern India. *Agricultural Economics Research Review*, 23(December), 265–272.

Kumar, M., Sharma, B. & Singh, O. (2009) Water saving and yield enhancing micro-irrigation technologies: how far can they contribute to water productivity in Indian agriculture? In: IWMI Books, Reports H042044. http://ageconsearch.umn.edu/handle/127972

Kumar, S., Singh, R., Asrey, R. & Nangare, D.D. (2012) Techno-economic evaluation of integrating canal water harvesting and drip irrigation for pomegranate production in a dry eco-region. *Irrigation and Drainage*, 61(3), 366–374. https://doi.org/10.1002/ird.658

Levidow, L., Zaccaria, D., Maia, R., Vivas, E., Todorovic, M. & Scardigno, A. (2014) Improving water-efficient irrigation: prospects and difficulties of innovative practices. *Agricultural Water Management*, 146, 84–94. https://doi.org/10.1016/j.agwat.2014.07.012

Lowenberg-DeBoer, J., Huang, I.Y., Grigoriadis, V. & Blackmore, S. (2020) Economics of robots and automation in field crop production. *Precision Agriculture*, 21(2), 278–299. https://doi.org/10.1007/s11119-019-09667-5

Luthra, S.K., Kaledonkar, M.J., Singh, O.P. & Tyagi, N.K. (1997) Design and development of an auto irrigation system. *Agricultural Water Management*, 33(2), 169–181. https://doi.org/10.1016/S0378-3774(96)01292-9

Madhava Chandran, K. & Surendran, U. (2016) Study on factors influencing the adoption of drip irrigation by farmers in humid tropical Kerala, India. *International Journal of Plant Production*, 10(3), 347–364. https://doi.org/10.22069/ijpp.2016.2902

Madramootoo, C.A. & Morrison, J. (2013) Advances and challenges with micro-irrigation. *Irrigation and Drainage*, 62(3), 255–261. https://doi.org/10.1002/ird.1704

Mahesh, R., Raja, N.A. & Archana, H.A. (2016) Performance of surface and subsurface drip fertigation on yield and water use efficiency of sugarcane. *2nd World Irrigation Forum, Thailand, November*, 6–8.

Mandal, K.G., Thakur, A.K. & Ambast, S.K. (2019) Current rice farming, water resources and micro-irrigation. *Current Science*, 116(4), 568–576. https://doi.org/10.18520/cs/v116/i4/568-576

Mishra, K.C. & Raghavendra, H.A. (2014) *A report on automated real time water and canal management*. Bangluru: Grievances Ministry of Personnel, Public Grievances & Pensions, Government of India.

Moench, M. (2003) Groundwater and food security in India. In: Prasad, K. (Ed.) *Water Resources and Sustainable Development: Challenges of 21st Century*. New Delhi, India: Shipra Publications, pp. 148–167.

Nagaraj, N. (2013) Rapporteurs' report: development of dryland agriculture: technological, institutional, infrastructural and policy imperatives. *Indian Journal of Agricultural Economics*, 68(3), 1–16.

Namara, R.E., Nagar, R.K. & Upadhyay, B. (2007) Economics, adoption determinants, and impacts of micro-irrigation technologies: empirical results from India. *Irrigation Science*, 25(3), 283–297. https://doi.org/10.1007/s00271-007-0065-0

Namara, R.E., Upadhyay, B. & Nagar, R.K. (2005) Adoption and impacts of microirrigation technologies empirical results from selected localities of Maharashtra and Gujarat States of India. In *Research Report 93*. https://doi.org/10.3910/2009.094

Narayana, V.V.D., Kalra, V.D. & Singh, O.P. (1980) Design of a farm layout for irrigation with limited discharges. *Agricultural Water Management*, 3(2), 143–151. https://doi.org/10.1016/0378-3774(80)90022-0

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Narayanamoorthy, A. (2007) Micro-irrigation and electricity consumption linkages in Indian agriculture: afield based study. In: International Conference on linkages between Energy and Water Management for Agriculture in Developing Countries. Hyderabad, India.

Naresh, R.K., Singh, S.P., Misra, A.K., Tomar, S.S., Kumar, P., Kumar, V. & Kumar, S. (2014) Evaluation of the laser leveled land leveling technology on crop yield and water use productivity in Western Uttar Pradesh. *African Journal of Agricultural Research*, 9(4), 473–478. https://doi.org/10.5897/ajar12.1741

National Committee on the Use of Plastics in Agriculture (NCPA). (1990) Status, potential and approach for adoption of drip and sprinkler irrigation systems. Pune, India.

National Rainfed Area Authority (NRAA). (2020) *Prioritization of districts for development planning in India, a composite index approach*. New Delhi, India: Task Force on Revisiting Prioritization of Rainfed Areas.

Osman, H.E.B. (2002) Evaluation of surface irrigation using gated pipes techniques in field crops and old horticultural farm. *Annals of Agricultural Science Cairo*, 47(2), 461–476.

Palanisami, K. & Raman, S. (2012) Potential and challanges in upscaling micro-irrigation in India Experiences from nine states. *Water Policy Research Highlight.*, 12, 383–384. https://doi.org/10.15290/sw.2012.12.37

Pandey, A.K., Singh, A.K., Singh, S.K. & Kumar, A. (2012) Improving strawberry productivity and quality through drip irrigation and polythene mulch in Bihar. *Environment and Ecology*, 30(3A), 760–763.

Panigrahi, B. & Panda, S.N. (2003) Optimal sizing of on-farm reservoirs for supplemental irrigation. *Journal of Irrigation and Drainage Engineering*, 129(2), 117–128. https://doi.org/10.1061/(ASCE)0733-9437(2003)129

Planning Commission. (2002) *Report of the steering committee on irrigation for the tenth five year plan (2002–2007)*. New Delhi, India: Government of India, p. 53.

Polak, P., Nanes, B. & Adhikari, D. (1997) A low cost drip irrigation system for small farmers in developing countries. *Journal of the American Water Resources Association*, 33(1), 119–124. https://doi.org/10.1111/j.1752-1688.1997.tb04088.x

Pramanik, S., Lai, S., Ray, R. & Patra, S.K. (2016) Effect of drip fertigation on yield, water use efficiency, and nutrients availability in Banana in West Bengal, India. *Communications in Soil Science and Plant Analysis*, 47(13–14), 1691–1700. https://doi.org/10.1080/00103624.2016.1206560

Rajput, T.B.S. & Patel, N. (2005) Enhancement of field water use efficiency in the Indo-Gangetic Plain of India. *Irrigation and Drainage*, 54(2), 189–203. https://doi.org/10.1002/ird.167

Raman, S. (2010) State-wise micro-irrigation potential in India - an assessment. Unpublished paper, Nature.

Ramanjaneyulu, A.N., Neelima, T.L. & Reddy, K.D. (2013) Role of farm ponds in rainfed Agriculture and research progress. Palem- 509215.

Rao, C.S., Lal, R., Prasad, V.N.S., Gopinath, K.A., Singh, R., Jakkula, V.S. *et al.* (2015) Potential and challenges of rainfed farming in India. *Advances in Agronomy*, 133, 115–181.

Ravichandran, V., Nayar, V. & Prakash, K. (2015) An evaluation of the SRI on increasing yield, water productivity and profitability; experiences from TN-IAMWARM Project. *Irrigation & Drainage Systems Engineering*, 04(02), 2–6. https://doi.org/10.4172/2168-9768.1000137

Ray, I. (2011) Farm-level incentives for irrigation efficiency: some lessons from an Indian canal. *Journal of Contemporary Water Research & Education*, 121(1), 64–71. http://ridum.umanizales.edu.co:8080/jspui/bitstream/6789/377/4/Muñoz_Zapata_Adriana_Patricia_Artículo_2011.pdf.

Reddy, J.M. (2013) Design of level-basin irrigation systems for robust performance. *Journal of Irrigation and Drainage Engineering*, 139(3), 254–260. https://doi.org/10.1061/(ASCE)IR.1943-4774.0000535

Reddy, K.Y., Reddy, L.N., Nagothu, S., Bhaskar, S. & Nakka, R. (2016) Sensors for water monitoring for improved on farm water management. 2 Nd World Irrigation Forum (WIF2), November, 1–8. http://www.icid.org/wif2_full_papers/wif2_w.3.1.07.pdf

Replogle, J.A. (1997) Costs of secondary parasitism in the facultative hyperparasitoid Pachycrepoideus dubius: Does host size matter? *Irrigation and Drainage Systems*, 11(3), 241–259. https://doi.org/10.1023/A, https://doi.org/10.1023/A:1005897812746

Rockström, J., Hatibu, N., Oweis, T. & Wani, S.P. (2007) Managing water in rain-fed agriculture. In: Molden, D. (Ed.) *Water for Food, Water for Life: a Comprehensive Assessment of Water Management in Agriculture*. Colombo, Sri Lanka: Earthscan, London, UK and International Water Management Institute (IWMI), pp. 315–348.

Rodell, M., Famiglietti, J.S., Wiese, D.N., Reager, J.T., Beaudoing, H.K., Landerer, F.W. & Lo, M.H. (2018) Emerging trends in global freshwater availability. *Nature*, 557(7707), 651–659. https://doi.org/10.1038/s41586-018-0123-1

Rosegrant, M., Ximing, C., Cline, S. & Nakagawa, N. (2002) The role of rain-fed agriculture in the future of global food production. EPTD Discussion Paper No. 90, Environment and Production Technology Division, IFPRI, Washington, DC, USA.

Safi, B., Neyshabouri, M.R., Nazemi, A.H., Massiha, S. & Mirlatifi, S.M. (2007) Water application uniformity of a subsurface drip irrigation system at various operating pressures and tape lengths. *Turkish Journal of Agriculture and Forestry*, 31(5), 275–285.

Samra, J.S. & Sharma, P.D. (2009) Food security - Indian scenario. *IPI-OUAT-IPNI* International Symposium on Potassium Role and Benefits in Improving Nutrient Management for Food Production, Quality and Reduced Environmental Damages, November 5-7., 15–43. https://doi.org/10.15957/j.cnki.jjdl.2009.07.004

Shah, T. (1993) In: Shah, T. (Ed.) *Groundwater markets and irrigation development: political economy and practical policy*. Bombay; New York: Oxford University Press. http://www.loc.gov/catdir/enhancements/fy0636/93244535-t.html

Shah, T. (2003) Groundwater management: synthesis of IWMI Research 1999–2003. Anand, India.

Shah, T. (2011) Past, present, and the future of canal irrigation in India. India Infrastructure Report 2011, 69–89. www.rimisp.org/wp-content/uploads/2010/05/Paper_Tushaar_Shah-4.pdf

Shah, T., Giordano, M. & Wang, J. (2004) Irrigation Institutions in a Dynamic Economy: What Is China Doing Differently from India? In Economic and Political Weekly, 93–112. International

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Symposium on Groundwater Sustainability (ISGWAS) Irrigation.

Sharda, R., Mahajan, G., Siag, M., Singh, A. & Chauhan, B.S. (2017) Performance of drip irrigated dry-seeded rice (Oryza sativa L.) in South Asia. *Paddy and Water Environment*, 15(1), 93–100. https://doi.org/10.1007/s10333-016-0531-5

Sharma, R.K. & Singh, H.R. (2009) Deep percolation losses in irrigation border-check. *Journal of Soil and Water Conservation*, 8(2), 65–74.

Sharma, S.K., Mishra, P.K., Panse, R. & Jamliya, G. (2018) Effect of irrigation methods on yields attributes and water productivity of wheat in vertisol of Betwa River Basin Commands of Vidisha District of M.P. *International Journal of Current Microbiology and Applied Sciences*, 7(08), 2670–2673. https://doi.org/10.20546/ijcmas.2018.708.277

Sharma, V., Singh, P.K., Bhakar, S.R., Yadav, K.K. & Lakhawat, S. (2019) Integration of soil moisture sensor based automated drip irrigation system for Okra crop. *International Journal of Pure & Applied Bioscience*, 7(4), 277–282. https://doi.org/10.18782/2320-7051.7642

Shiva Shankar, M., Ramanjaneyulu, A.N., Neelima, T.L. & Das, A. (2015) Sprinkler irrigation – an asset in water scarce and undulating areas. In: Rajkhowa, D.J., Das, A., Ngachan, S.V., Sikka, A.K. & Lyngdoh, M. (Eds.) *Integrated Soil and Water Resource Management for Livelihood & Environmental Security (Issue June, pp. 259–283)*. Umiam, Meghalaya, India: ICAR Research Complex for NEH Region.

Siebert, S., Burke, J., Faures, J.M., Frenken, K., Hoogeveen, J., Doll, P. & Portmann, F.T. (2010) Groundwater use for irrigation – a global inventory. *Hydrology and Earth System Sciences*, 14(10), 1863–1880. https://doi.org/10.5194/hess-14-1863-2010

Singh, N.T. (2005) *Irrigation and soil salinity in the Indian subcontinent: past and present*. Bethlehem, USA: Lehigh University Press. https://lupress.cas.lehigh.edu/content/irrigation-and-soil-salinity-indian-subcontinent-past-and-present

Sivanappan, R.K., Rao, A.S. & Dikshit, N.K. (1994) *Drip irrigation in India*. New Delhi, India: Indian National Committee on Irrigation and Drainage.

Sood, R., Kaur, M. & Lenka, H. (2013) Design and development of automatic water flowmeter. *International Journal of Computer Science, Engineering and Applications*, 3(3), 49–59. https://doi.org/10.5121/ijcsea.2013.3306

Souza, C.F. & Bizari, D.R. (2018) Soil solution distribution in subsurface drip irrigation in sugarcane. *Engenharia Agrícola*, 38(2), 217–224. https://doi.org/10.1590/1809-4430-eng.agric.v38n2p217-224/2018

Srivastava, R.C., Mohanty, S., Singandhuppe, R.B., Mohanty, R.K., Behera, M.S., Ray, I.P.L. & Sahoo, D. (2010) Feasibility evaluation of pressurized irrigation in canal commands. *Water Resources Management*, 24(12), 3017–3032. https://doi.org/10.1007/s11269-010-9592-1

Srivastava, R.C. & Upadhayaya, A. (1998) Study on feasibility of drip irrigation in shallow ground water zones of eastern India. *Agricultural Water Management*, 36(1), 71–83. https://doi.org/10.1016/S0378-3774(97)00046-2

Sudharshan, N., Karthik, A., Kiran, J.S.S. & Geetha, S. (2019) Renewable energy based smart irrigation system. *Procedia Computer Science*, 165, 615–623. https://doi.org/10.1016/j.procs.2020.01.055

Surendran, U., Sushanth, C.M., Mammen, G. & Joseph, E.J. (2015) Modelling the crop water requirement using FAO-CROPWAT and assessment of water resources for sustainable water resource management: a case study in Palakkad district of humid tropical Kerala, India. *Aquatic Procedia*, 4, 1211–1219. https://doi.org/10.1016/j.aqpro.2015.02.154

Suresh, A., Aditya, K.S., Jha, G. & Pal, S. (2019) Micro-irrigation development in India: an analysis of distributional pattern and potential correlates. *International Journal of Water Resources Development*, 35(6), 999–1014. https://doi.org/10.1080/07900627.2018.1504755

Suresh, A. & Samuel, M.P. (2020) Micro-irrigation development in India: challenges and strategies. *Current Science*, 118(8), 1163–1168.

Thakkar, H. (1999) Assessment of irrigation in India (Draft #1; Thematic Review IV.2). New Delhi, India.

Tyagi, N.K. (1997) Salinity management: the CSSRI experience and future research agenda. In: Snellen, W.B. (Ed.) *Towards integration of irrigation and drainage management*. Wageningen, the Netherlands: ILRI, pp. 17–27.

Tyagi, N.K., Kaushal, R.K., Ram, S. & Sampath, R.K. (2012) Towards determining the optimal size of unit irrigation command area. *International Journal of Water Resources Development, Routledge*, 9(4), 425–438.

Velasco-Muñoz, J.F., Aznar-Sánchez, J.A., Batlles-delaFuente, A. & Fidelibus, M.D. (2019) Sustainable irrigation in agriculture: an analysis of global research. *Water*, 11(9), 1758. https://doi.org/10.3390/w11091758

Vij, A., Singh, V., Jain, A., Bajaj, S., Bassi, A. & Sharma, A. (2020) ScienceDirect IoT and machine learning approaches for automation of farm IoT and machine approaches automation of farm on computational intelligence for IoT and machine learning approaches for automation of farm irrigation systems. *Procedia Computer Science*, 167, 1250–1257. https://doi.org/10.1016/j.procs.2020.03.440

Walker, W.R. (1989) Guidelines for designing and evaluating surface irrigation systems. Food and Agriculture Organization of the United Nations (FAO). FAO Irrigation and Drainage Paper No. 45. Rome, Italy.

Wani, S.P., Jakkula, V.S. & Singh, D. (eds.) (2017) Doubling farmers' income: KISAN–MITrA. Proceedings of National Workshop on Doubling Farmers' Income through Scaling up: KISAN–MITrA (Knowledge-based Integrated Sustainable Agriculture Network – Mission India for Transforming Agriculture) Patancheru 502 324. Telangana, India: International Crops Research Institute for the Semi-Arid Tropics. 160 pp. ISBN No. 978–92–9066-594-6 Order code: CPE175.

Wani, S.P., Pathak, P., Jangawad, L.S., Eswaran, H. & Singh, P. (2003) Improved management of vertisols in the semi-arid tropics for increased productivity and soil carbon sequestration. *Soil Use and Management*, 19(3), 217–222. https://doi.org/10.1111/j.1475-2743.2003.tb00307.x

Wani, S.P., Rockstorm, J. & Oweis, T. (2009) *Rainfed agriculture: unlocking the potential*. Wallingford, United Kingdom: CAB International 10.1079/9781845933890.0000.

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Yadav, S., Gill, G., Kukal, S. S., Humphreys, E., Rangarajan, R. & Walia, U.S. (2010) Water balance in dry seeded and puddled transplanted rice in Punjab, India. 19th World Congress of Soil Science, Soil Solutions for a Changing World, 1-6th August, 2008–2011.

Yang, M., Guan, X., Liu, Y., Cui, J., Ding, C., Wang, J. *et al.* (2019) Effects of drip irrigation pattern and water regulation on the accumulation and allocation of dry matter and nitrogen, and water use efficiency in summer maize. *Acta Agronomica Sinica*, 45(3), 443–459. https://doi.org/10.3724/sp.j.1006.2019.83026

How to cite this article: Gupta, A., Singh, R.k., Kumar, M., Sawant, C.P. & Gaikwad, B.B. (2022) On-farm irrigation water management in India: Challenges and research gaps. *Irrigation and Drainage*, 71(1), 3–22. Available from: https://doi.org/10.1002/ird.2637

15310361, 2022, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/ird.2637 by Amrita Vishwa Vidyapeetham Coimbatore, Wiley Online Library on [18/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License